

SLIDE 1 — Introduction

Slide content

- Traffic camera generates continuous video
- Cloud processing → latency + bandwidth + privacy concerns
- Question: What if AI could process data locally?

YOU — ~20 sec

Imagine a traffic camera detecting a vehicle, but instead of making the decision immediately, it has to send that entire video to a distant data center, wait for the AI to process it, and then receive the result.

That takes time, uses bandwidth, and also means that our data has to leave the place where it was generated.

So, what if we could bring AI closer to the data itself?

That is where Edge AI comes in. Good morning everyone. Today, we are going to talk about Edge AI — its working, applications, advantages, challenges, and its future.

SLIDE 2 — Why did Edge AI emerge?

Slide content

- Traditional computing → processing on device
- Cloud AI → powerful remote processing
- Problems: latency ■ bandwidth ■ privacy ■ connectivity
- Edge computing → computation moves closer to data
- Edge AI → AI inference at/near the data source

YOU — ~35 sec

Initially, computing and data processing were mostly done locally. Then cloud computing became popular, where large amounts of data could be sent to powerful remote servers for processing.

This made AI much more powerful, but applications such as autonomous vehicles, surveillance and industrial machines often need decisions within milliseconds.

Sending all of their data to the cloud can introduce latency, consume bandwidth and create privacy concerns.

So computation started moving closer to where the data is produced. When artificial intelligence is performed at this edge, we get Edge AI.

SLIDE 3 — What exactly is Edge AI?

Slide content

- Definition: AI processing performed on or near the device where data is generated
- Examples: smartphone ■ smart camera ■ vehicle ■ industrial machine
- Edge ≠ no cloud
- Edge → fast local inference
- Cloud → training ■ storage ■ analytics ■ model management

YOU — ~35 sec

So, what exactly is Edge AI? In simple terms, Edge AI means running artificial intelligence models on the device itself, or on a nearby edge device, instead of sending all the data to a distant cloud server.

For example, a smartphone can recognize your face locally, a security camera can detect a person locally, and a vehicle can process information from its cameras and sensors without continuously depending on the cloud.

The important point is not that the cloud disappears. Rather, the edge handles time-sensitive processing, while the cloud can still be used for storage, training, analytics and other heavier tasks.

Handoff: "Now that we understand what Edge AI is and why it emerged, let's look at the system components and then follow the data through the complete workflow. I'll hand it over to [Name]."

SLIDE 4 — Components of Edge AI — Chips & Model Optimization

1. Powerful AI Chips

Modern devices now include dedicated AI processors called NPUs (Neural Processing Units).

Examples:

- Apple Neural Engine
- Google Tensor Chip
- Qualcomm AI Engine
- NVIDIA Jetson platform

These chips can process AI tasks much faster while consuming less power.

2. Model Optimization

Researchers developed techniques to make AI models smaller and faster. Some important techniques are:

- **Quantization** — Reduces the precision of model weights, making models smaller and faster.
- **Pruning** — Removes unnecessary neurons or connections from a neural network.
- **Knowledge Distillation** — Trains a smaller model to learn from a larger, more complex model.

These methods allow advanced AI models to run efficiently even on smartphones and IoT devices.

SLIDE 5 — Components of Edge AI — TinyML, IoT & Networks

3. TinyML

TinyML enables AI models to run on small, low-power microcontrollers, making smart sensors and wearable devices more intelligent and energy-efficient.

4. Growth of IoT and 5G

The rapid growth of IoT devices has increased the amount of data generated by smart sensors, cameras, and connected devices, making local processing with Edge AI more efficient than sending everything to the cloud.

5. Faster Networks

Finally, the advancements in 4G, 5G, Wi-Fi 6, and fiber broadband have enabled faster communication and supported real-time Edge AI applications such as autonomous vehicles, smart cities, and industrial automation.

SLIDE 6 — Basic Working of an Edge AI System

Slide content

- **Hardware** — cameras, microphones, sensors, IoT devices (data collection) + edge device (smartphone, smart camera, drone, robot, embedded system)
- **Software** — trained AI model analyzes data and makes decisions locally, run on CPU / GPU / NPU (NPU = specialized for AI processing)
- **Network & Cloud** — device connects via Wi-Fi / 4G / 5G; cloud used for training, storage and model updates

SPEAKER

An Edge AI system basically has three major parts.

First is the hardware — consist of cameras, microphones, sensors, and IoT devices that collect the data, then this data is processed by an edge device, such as a smartphone, smart camera, drone, robot, or any embedded system.

Second is the software — Inside the device, a trained AI model analyzes the data and makes decisions locally. To perform these AI tasks efficiently, modern devices use specialized hardware like CPUs, GPUs, and NPUs, where NPUs are specifically designed for AI processing.

And finally, there is the network and cloud layer.

The device can communicate through Wi-Fi, 4G, or 5G, while the cloud is mainly used for training AI models, storing data, and delivering model updates.

SLIDE 7 — Current Applications of Edge AI

1. Smartphones

- Face Unlock
- Voice assistants
- AI-powered photo editing

Instead of uploading every photo to the cloud, the phone processes many tasks locally, making them faster and more private.

2. Autonomous Vehicles

- Object Detection
- Lane Detection
- Traffic Sign Recognition

3. Healthcare

- Smartwatches
- ECG Monitoring
- Portable Diagnostics

4. Smart Homes & Surveillance

- Smart Speakers
- Smart Doorbells
- Fire & Intruder Detection

SLIDE 8 — Advantages of Edge AI

- Low latency — faster decisions close to the source
- Lower bandwidth — less raw data needs to travel to the cloud
- Privacy potential — sensitive data can remain local
- Resilience — can continue operating with weak/intermittent connectivity
- Reduced transmission cost — less continuous cloud data transfer
- Real-time intelligence — useful for safety- and time-critical applications

PERSON 3 — ~30 sec

Edge AI has several important advantages. It provides low latency, because decisions can happen near the source.

It can reduce bandwidth usage by avoiding continuous transmission of raw data.

It also offers better privacy potential because sensitive information can remain local, and it can be more resilient when connectivity is unreliable.

These properties make Edge AI especially useful for real-time applications.

SLIDE 9 — Limitations & Challenges

- Limited compute & memory — edge devices are resource-constrained
- Energy consumption — important for battery-powered devices
- Security — many distributed devices increase the attack surface
- Accuracy vs. efficiency — optimization can introduce accuracy loss
- Model updates — deploying new models across many devices is difficult
- Fleet management — monitoring, debugging and maintenance at scale

PERSON 3 — ~30 sec

However, Edge AI introduces its own challenges. Edge devices have limited compute, memory and energy compared with cloud servers.

Security is also important because many distributed devices can increase the attack surface.

There is a constant trade-off between model accuracy and efficiency, and updating or debugging thousands of deployed devices can be difficult.

Handoff: "So Edge AI solves important problems, but it also creates new engineering challenges. Now let's look at where this technology is heading."

SLIDE 10 — Future of Edge AI + Conclusion

Future — point format

- Autonomous systems — smarter vehicles, robots and drones
- Industrial Edge AI — predictive maintenance and real-time quality inspection
- Smart cities — adaptive traffic and intelligent infrastructure
- On-device AI — increasingly capable AI directly on phones and personal devices
- Edge + 5G/6G — faster communication between devices and nearby compute
- TinyML — AI on very small, low-power embedded devices

Conclusion — point format

- Edge: fast, local, real-time decisions
- Cloud: large-scale training, storage and analytics
- Future: Edge + Cloud working together

PERSON 4 — ~35 sec

Looking ahead, Edge AI can expand into autonomous systems, industrial automation, smart cities, healthcare and increasingly capable AI on personal devices.

Technologies such as 5G and 6G, specialized AI hardware and TinyML can make local intelligence even more practical.

To conclude, Edge AI brings intelligence closer to where data is generated. The edge handles fast local decisions, while the cloud handles large-scale training, storage and analytics.

So the future is not necessarily Edge versus Cloud. It is Edge working together with Cloud.

Thank you.